

 Open in New Tab

To print the PDF document, please use the Print button on the toolbar.



Meta Review Status: The meta review and decision have not been released yet.



AI Review from DeepReviewer will be automatically processed



Summary

This paper introduces a hybrid framework for enhancing algorithmic creativity in text generation, combining multi-token teacherless prediction (MTP) and discrete diffusion methods (SEDD) with traditional next-token prediction (NTP). The authors propose seed-conditioning and

temperature scaling as mechanisms to control the generation process, aiming to balance coherence, diversity, and memorization. The core idea is to move beyond the limitations of NTP, which often struggles with generating novel outputs, by integrating MTP and SEDD, which offer more flexibility in exploring the output space. The paper's methodology involves defining a coherence loss (L_{coh}), a creativity loss (L_{cre}), and a mixing loss (L_{mix}) to guide the generation process. The coherence loss encourages the model to produce outputs that are semantically consistent with the input, while the creativity loss promotes the generation of novel and diverse outputs. The mixing loss combines these two objectives, allowing the model to balance coherence and creativity. Seed-conditioning is used to introduce a starting point for the generation, while temperature scaling controls the randomness of the output. The experimental evaluation is conducted on a synthetic dataset comprising 200 examples, each associated with one of four minimal combinatorial tasks: Sibling Discovery, Triangle Discovery, Circle Construction, and Line Construction. The results indicate that MTP and SEDD outperform NTP in terms of diversity and reducing memorization, albeit with a slight cost to coherence. The authors conclude that their hybrid framework offers a scalable approach for complex problem domains, suggesting that the combination of NTP, MTP, and SEDD provides a more effective way to generate creative text than relying on NTP alone. However, the paper's reliance on synthetic data and the lack of detailed implementation information limit the generalizability and reproducibility of the findings. Overall, the paper presents an interesting approach to enhancing creativity in text generation, but it requires further validation and clarification to fully realize its potential.

✓ Strengths

The paper's primary strength lies in its novel approach to enhancing algorithmic creativity in text generation. The proposed hybrid framework, which combines next-token prediction (NTP) with multi-token teacherless prediction (MTP) and discrete diffusion methods (SEDD), is a creative and potentially effective way to address the limitations of traditional NTP. The introduction of seed-conditioning and temperature scaling as mechanisms to control the generation process is also a valuable contribution, providing a means to balance coherence, diversity, and memorization. The authors' attempt to formalize the trade-offs between coherence and creativity through the use of distinct loss functions (L_{coh} , L_{cre} , and L_{mix}) is a notable effort, even if the specifics of these functions could be clearer. The paper's focus on reducing memorization while increasing diversity is particularly relevant in the context of creative text generation, where the ability to generate novel outputs is crucial. The experimental results, while limited to a synthetic dataset, do provide some evidence that MTP and SEDD can outperform NTP in terms of diversity and reducing memorization. The paper's exploration of different methods for text generation, and the attempt to combine them in a hybrid framework, is a valuable contribution to the field. The authors' recognition of the limitations of NTP and their attempt to address these limitations through the use of MTP and SEDD is a significant step forward. The paper also provides a clear motivation for the research, highlighting the need for more creative text generation methods. The overall structure of the paper is logical, with a clear introduction, methodology, experimental setup, and results sections. The authors' attempt to address a significant challenge in text generation, namely the balance between coherence and creativity, is commendable. The paper's exploration of different methods for text generation, and the attempt to combine them in a hybrid framework, is a valuable contribution to the field.

✗ Weaknesses

After a thorough examination of the paper, several significant weaknesses have emerged, which I will detail with supporting evidence. Firstly, the paper suffers from a lack of clarity in its presentation, particularly regarding the definition of key symbols and terms. For instance, in Equation 1, the variables 'x' and 'y' are used without explicit definition within the equation itself, although they are mentioned in the surrounding text. Similarly, in Equation 4, the prompts P_r and P_a are introduced without a clear explanation of their specific roles and why these particular notations are used. The symbol '#' on line 067 is used without a clear definition of what it represents, and the diversity metric, represented by #, lacks a detailed explanation of how it is calculated and why it doesn't reach 100% when memorization is zero. The latent representation Z_t in Equation 6 is introduced without a clear explanation of its derivation or meaning within the diffusion process. These ambiguities make it difficult to fully understand the proposed methodology and replicate the results. The lack of precise definitions for key variables and symbols makes it extremely difficult to follow the proposed methodology. For instance, in Equation 1, the variables 'x' and 'y' should be explicitly defined as the input and output sequences, respectively, and the specific loss function used should be named (e.g., cross-entropy). The symbol '#' needs a clear definition, including its units (number of tokens, words, or characters). The diversity metric, represented by #, needs a more detailed explanation of how it is calculated and why it doesn't reach 100% when memorization is zero. The prompts P_r and P_a in Equation 4 should be

clearly defined as restrictive and adaptive prompts, respectively, and the rationale for using these specific notations should be provided. The latent representation Z_t in Equation 6 needs a more detailed explanation of how it is derived and what it represents in the context of the diffusion process. The authors should also clarify the meaning of p_t and correct the typo in "temperature". The reference to "Table below" is confusing and should be replaced with a direct placement of the table after the paragraph. These are just a few examples, and the authors should carefully review the entire manuscript to ensure that all symbols, variables, and concepts are clearly defined and explained. This lack of clarity significantly hinders the paper's readability and makes it difficult to assess the validity of the proposed approach. My confidence in this assessment is high, as these issues are directly observable in the paper's text and equations. Secondly, the experimental validation is severely limited by the use of a small synthetic dataset. The paper explicitly states that the experiments were conducted on a synthetic dataset comprising 200 examples, each associated with one of four minimal combinatorial tasks. While synthetic data can be useful for initial testing, it is crucial to validate the proposed method on larger, real-world datasets to demonstrate its practical applicability and generalizability. The current experiments do not provide sufficient evidence to support the claims made in the paper. The authors should consider using standard benchmark datasets for text generation tasks, such as those used in summarization, dialogue, or story generation. This would allow for a more meaningful comparison with existing methods and provide a better understanding of the strengths and weaknesses of the proposed approach. The use of a small synthetic dataset limits the generalizability of the findings and raises concerns about the practical applicability of the proposed method. My confidence in this assessment is high, as the paper explicitly states the use of a synthetic dataset with a limited number of examples. Thirdly, the paper lacks sufficient details about the implementation, making it difficult to reproduce the results. The paper mentions the use of a standard transformer architecture but does not specify the number of layers, hidden size, attention mechanism details, or other architectural specifics. While some hyperparameters are mentioned, a comprehensive description of all relevant hyperparameters is missing. The lack of detailed implementation information hinders the reproducibility of the work. My confidence in this assessment is high, as the paper lacks specific details about the transformer architecture and comprehensive hyperparameter settings. Fourthly, the paper's experimental validation is limited to synthetic and minimal tasks, which may not fully represent real-world scenarios. The combinatorial tasks, while useful for initial testing, lack the complexity and nuance of real-world text generation. For example, the tasks do not capture the long-range dependencies and semantic coherence required in tasks like story generation or complex reasoning. This raises concerns about the generalizability of the findings to more complex and diverse datasets. The evaluation should include tasks that require more sophisticated understanding of context and world knowledge. My confidence in this assessment is high, as the paper explicitly describes the use of minimal combinatorial tasks. Fifthly, the study could benefit from a more extensive comparison with a broader range of existing methods and models in the field. While the paper compares against NTP, MTP, and SEDD, it lacks a thorough comparison with other state-of-the-art text generation models, particularly those that incorporate different decoding strategies or training objectives. This makes it difficult to contextualize the performance of the proposed method relative to the broader landscape of text generation techniques. A more comprehensive comparison would provide a clearer understanding of the advantages and disadvantages of the proposed approach. My confidence in this assessment is high, as the paper's experimental section primarily focuses on comparing the proposed variations of NTP, MTP, and SEDD. Finally, the paper could provide more details on the computational complexity and scalability of the proposed methods, especially the discrete diffusion model (SEDD). The paper does not provide a detailed analysis of the computational cost associated with the iterative denoising process of SEDD, including the time and memory requirements for training and inference. This lack of information makes it difficult to assess the practical applicability of the method, especially for large-scale text generation tasks. A thorough analysis of computational resources is crucial for evaluating the feasibility of the proposed approach. My confidence in this assessment is high, as the paper lacks any discussion or empirical evaluation of the computational cost associated with the proposed methods.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors must significantly improve the clarity of their writing by providing explicit definitions for all symbols and terms used in the paper. This includes defining 'x' and 'y' in Equation 1, explaining the meaning of '#' on line 067, clarifying the calculation of the diversity metric, providing a rationale for the notation P_r and P_a in Equation 4, and explaining the derivation and meaning of Z_t in Equation 6. The authors should also clarify the meaning of p_t and correct the typo in "temperature". The reference to "Table below" should be replaced with a direct placement of the table after the paragraph. The authors should carefully review the entire manuscript to ensure that all symbols, variables, and concepts are clearly defined and explained. This will significantly improve the paper's readability and allow for a more thorough assessment of the proposed methodology. Secondly, the authors should conduct experiments on larger, real-world datasets to demonstrate the practical applicability and generalizability of their method. This could include standard benchmark datasets for text generation tasks, such as those used in summarization, dialogue, or story generation. The use of real-world datasets will provide a more meaningful evaluation of the proposed method and allow for a more direct comparison with

existing approaches. Thirdly, the authors should provide a more detailed description of the model architecture, including the number of layers, the size of the hidden states, and the type of activation functions used. The hyperparameter settings, such as the learning rate, batch size, and optimization algorithm, should also be clearly specified to ensure the reproducibility of the results. This will allow other researchers to replicate the findings and build upon the proposed method. Fourthly, the authors should consider evaluating their methods on more complex and realistic datasets. This could include datasets from tasks such as story generation, where the model needs to maintain coherence over longer sequences, or question answering, where the model needs to demonstrate reasoning abilities. Additionally, the authors should explore the use of metrics that capture the semantic quality and diversity of the generated text, beyond simple n-gram based metrics. For example, metrics based on semantic similarity or entailment could provide a more nuanced evaluation of the creative output. Furthermore, the authors should provide a more detailed analysis of the failure cases of their method, identifying the specific types of tasks or inputs where the method struggles. This would provide valuable insights into the limitations of the approach and guide future research directions. The inclusion of human evaluation would also be beneficial to assess the perceived creativity of the generated outputs. Fifthly, to strengthen the comparison with existing methods, the authors should include a more comprehensive set of baselines in their experiments. This should include not only different decoding strategies but also different model architectures and training objectives. For example, comparing against models trained with reinforcement learning or adversarial training could provide a more complete picture of the performance of the proposed method. Additionally, the authors should provide a more detailed analysis of the strengths and weaknesses of each method, identifying the specific scenarios where each method performs best. This would help to contextualize the performance of the proposed method and provide a more nuanced understanding of its advantages and disadvantages. The authors should also consider using ablation studies to analyze the impact of different components of their method, such as the seed-conditioning and temperature scaling, on the overall performance. This would help to understand the contribution of each component to the overall performance of the framework. Finally, to address the concerns about computational complexity, the authors should provide a detailed analysis of the time and memory requirements of their method, including a breakdown of the computational cost of each step. This should include an analysis of the scalability of the method to larger datasets and more complex models. The authors should also explore techniques for reducing the computational cost of their method, such as using more efficient sampling algorithms or model compression techniques. Furthermore, the authors should provide a comparison of the computational cost of their method with other state-of-the-art text generation methods, providing a clear understanding of the trade-offs between performance and computational cost. This would help to assess the practical applicability of the method and guide future research in this area.

? Questions

Several key questions arise from my analysis of this paper. Firstly, how well does the proposed method generalize to more complex and diverse datasets beyond the minimal tasks used in the study? The current experiments are limited to a small synthetic dataset, and it is unclear whether the findings would hold true for real-world text generation tasks. Secondly, can the authors provide more insights into the computational efficiency and scalability of the discrete diffusion model (SEDD), especially when applied to large-scale text generation tasks? The paper lacks a detailed analysis of the computational cost associated with the iterative denoising process of SEDD, making it difficult to assess its practical applicability. Thirdly, how does the proposed method compare to other state-of-the-art approaches in terms of performance and computational cost? The paper primarily compares against NTP, MTP, and SEDD, but it lacks a thorough comparison with other existing methods, particularly those that incorporate different decoding strategies or training objectives. Fourthly, what is the specific definition of the coherence loss (L_{coh}), creativity loss (L_{cre}), and mixing loss (L_{mix}) used in the paper? The paper mentions these loss functions but does not provide a clear mathematical definition of each. Fifthly, what is the rationale behind the specific choice of hyperparameters used in the experiments? The paper mentions some hyperparameters but does not provide a detailed explanation of why these specific values were chosen. Finally, what are the limitations of the proposed method, and what are the potential directions for future research? The paper does not provide a thorough discussion of the limitations of the proposed approach, and it would be beneficial to have a more detailed discussion of the potential avenues for future research.



Scores

Soundness: 1.75

Presentation: 1.5

Contribution: 1.75

Rating: 3.5

 AI Review from ZGCA will be automatically processed Summary

The paper studies "algorithmic creativity" in text generation on four minimal combinatorial tasks (Sibling Discovery, Triangle Discovery, Circle Construction, Line Construction). It compares Next-Token Prediction (NTP) to Multi-Token Teacherless Prediction (MTP) and a discrete diffusion model (SEDD), and introduces seed-conditioning (input-level noise) and temperature scaling to trade off coherence, diversity, and memorization. Creativity is measured via: coherence as the fraction of outputs that satisfy task-specific validity rules (Sections 1, 5), diversity as the fraction of unique outputs, and memorization as exact replication of a training output. The method also proposes an alignment loss $L_{\text{mix}} = 1 - \cos(E(\text{Pr}), E(\text{Pa}))$ to align restrictive and adaptive prompts (Sections 1, 4), and a noise schedule $p_{\text{noise}} = \min(0.9, \alpha T)$ with method-specific α (Section 4). On a synthetic dataset of 200 examples (Section 5), results claim that MTP and SEDD, particularly with nonzero temperature and seed-conditioning, reduce exact memorization from near 100% to near 0% and increase diversity to ~ 1.00 while maintaining 50–80% coherence (Table 1; Section 6 tables). Ablations vary temperature, seed length, and token ordering (Sections 5–6).

 Strengths

- Clear, simple metrics and objective for the target trade-offs (coherence, diversity, exact memorization) with explicit formulas (Sections 1–2, 5).
- Systematic comparison across NTP, MTP, and discrete diffusion (SEDD) under controlled decoding conditions including temperature and seed-conditioning (Sections 1, 4–6).
- Transparent reporting of key control knobs, e.g., $p_{\text{noise}} = \min(0.9, \alpha T)$ with α per method; use of $L_{\text{total}} = L_{\text{gen}} + \lambda L_{\text{mix}}$ with λ specified; number of diffusion steps (10) (Sections 4–5).
- Ablations over temperatures ($T = 0, 0.5, 1.0, 2.0$), seed-conditioning on/off, and seed lengths; qualitative discussion of trade-offs (Sections 5–7).
- Results are consistent in direction: adding controlled stochasticity and seed-conditioning reduces exact memorization and increases diversity (Sections 6–7).

✗ Weaknesses

- Experimental scope is very limited: a 200-example synthetic dataset across toy tasks with rule-based validity (Section 5). This constrains external validity.
- Methodological under-specification hampers reproducibility: MTP's training objective and architecture details are not concretely described; SEDD's discrete diffusion loss L_{diff} and the "absorb variant" are not fully specified; model sizes and training hyperparameters are only partially reported (Section 4–5).
- Statistical rigor is insufficient. Claims of "statistical significance" rely on consistency over runs but no variance estimates, confidence intervals, or hypothesis tests are provided; all tables present point estimates (Section 6; Section 7's assertion of significance).
- Memorization is measured only as exact output match (Sections 5–6), which is a weak criterion; it misses partial or structural memorization and may be trivially reduced by noise.
- Diversity is raw uniqueness ($D = |\text{unique}|/\text{total}$) (Sections 1–2, 5), which can be artificially inflated by stochasticity without guaranteeing meaningful variety.
- Coherence is defined as rule satisfaction on synthetic patterns (e.g., count of arrows, presence of "tri:" prefix) (Section 5), which may not correlate with linguistic or semantic coherence in real tasks.
- Baselines are incomplete. Common decoding baselines (top-k/top-p, contrastive decoding, bounded entropy/top-h) are not included, despite relevance to creativity-diversity trade-offs.
- The α constants for p_{noise} and the choice of λ are presented but not justified by principled selection or sensitivity analysis beyond coarse ablations (Sections 4–6).
- No real-world or standard benchmarks are included; claims of scalability remain speculative (Sections 1, 7).
- Resource transparency is incomplete: code/artefacts are not provided; some seeds are fixed but key seeds (initialization) and training details (batch sizes, schedules) are omitted (Section 5).

? Questions

- Please specify the MTP objective and training procedure: What loss replaces standard teacher-forced NLL? How are multi-token predictions parameterized, and how is teacherlessness enforced during training?
- For SEDD, please formalize L_{diff} and the discrete diffusion process: What corruption/denoising distributions are used, what is the "absorb variant" concretely, and how are gradients estimated over discrete states?
- What are the model sizes and training details (parameters, depth/width, tokenizer, training steps/epochs, batch sizes, optimizer, learning rate schedule, regularization) for NTP vs MTP vs SEDD? How is compute matched across methods?
- How were α values in $p_{\text{noise}} = \min(0.9, \alpha T)$ chosen (0.3/0.5/0.7)? Please provide sensitivity analyses beyond the reported temperatures.
- Could you report standard deviations/confidence intervals over seeds and runs, and perform statistical tests to support claims of significance (e.g., on coherence and memorization differences)?
- How is the dataset split to prevent leakage between training and evaluation prompts/targets? Are the 200 examples all seen during training, or is there a held-out test set? If there is no held-out set, how should we interpret memorization and coherence?
- Please expand the memorization analysis: report n-gram overlap, edit distance, structural similarity (e.g., pattern-level matches) to detect partial memorization.
- How robust are the findings to stronger decoding baselines (top-k, nucleus/top-p, contrastive decoding, bounded-entropy/top-h)?
- Can you provide results on at least one standard text generation task (e.g., summarization or data-to-text) with both automatic and human evaluation, to evaluate external validity of the creativity-diversity-coherence trade-offs?
- What is L_{mix} 's impact? Please show ablations over λ , and report whether L_{mix} improves coherence without harming diversity compared to $\lambda=0$.

- Please clarify the counts in Section 5 vs Section 6: at one point, you mention 20 examples × 5 outputs per example, yet later you aggregate over 100 runs per condition. How many unique prompts, runs, and outputs underpin each table?

Limitations

- The study is restricted to synthetic, rule-based tasks with limited semantic depth (Section 5), limiting generalization to real-world language generation.
- Memorization is measured only via exact string match (Sections 5–6), overlooking partial or structural memorization and potentially overstating gains.
- Diversity as raw uniqueness can be inflated by noise; lack of semantic diversity metrics or human judgments makes it hard to assess creative quality (Sections 5–7).
- Methodological under-specification (MTP/SEDD objectives, diffusion specifics) reduces reproducibility and may mask implementation-sensitive effects (Sections 4–5).
- No evaluation on standard benchmarks or human evaluation, so claims of scalability and practical utility remain speculative (Sections 1, 7).
- Potential negative impact: overemphasis on stochasticity could degrade coherence in user-facing applications; without robust safeguards, higher diversity could increase hallucinations or non-compliant outputs.

Image Evaluation

Cross-Modal Consistency: 31/50

Textual Logical Soundness: 18/30

Visual Aesthetics & Clarity: 13/20

Overall Score: 62/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

- Major 1: Inconsistent table numbering and references (“Table below” used multiple times) blocks reliable mapping across text and visuals. Evidence: “To further illustrate... Table below summarizes” (Sec. 4, Methods).
- Major 2: Diversity units inconsistent across tables (percentages, raw numbers, and fractions 0.20/1.00). Evidence: “Diversity (%)” (Sec. 2 table) vs “Diversity” with 0.20/1.00 (Sec. 4 table).
- Major 3: Sample size inconsistency between setup and results (“100 runs per condition” vs 20×5 outputs total). Evidence: “over 20... 5 outputs” (Sec. 5) vs “over 100 generation runs per condition” (Sec. 6).
- Minor 1: Redundant summary tables (Table 1 and Table 2) with near-identical numbers may confuse which is authoritative. Evidence: “Table 1... performance metrics” (Intro) vs “Table 2: Comparison...” (Sec. 3).

2. Text Logic

- Major 1: Claims of “rigorous statistical analyses” without error bars, CIs, or tests; no method for significance. Evidence: “extensive experimentation and rigorous statistical analyses” (Abstract).

- Major 2: Two-stage pipeline description unclear: “first NTP base then seed-conditioning,” yet MTP/SEDD evaluated standalone. Evidence: “First, we generate a base sequence using conventional NTP...” (Sec. 4).
- Minor 1: Alignment loss L_{mix} is introduced and λ fixed, but no ablation or demonstrated effect. Evidence: “we employ an alignment loss... L_{mix} ” (Sec. 4); “The final setup employed $\lambda=0.5$ ” (Sec. 5).

3. Figure Quality

- Major 1: Multiple unnumbered tables (“Table below”) lack captions/IDs; impedes cross-referencing and figure-alone comprehension. Evidence: “A summary of the evaluation metrics is provided in Table below.” (Sec. 5, Sec. 6).
- Minor 1: Mixed units for Diversity and occasional missing percent signs reduce immediate interpretability. Evidence: “Diversity” entries like 69/84/100 (Sec. 6 table, no “%”).

Key strengths:

- Clear problem framing of creativity–memorization–diversity trade-offs with simple, reproducible synthetic tasks.
- Consistent directional findings: MTP/SEDD reduce memorization to ~0–1% and raise diversity near 1.00 under $T>0$.
- Practical levers (temperature, seed-conditioning) are specified and repeatable; α values and λ reported.

Key weaknesses:

- Cross-referencing and unit inconsistencies across tables hinder verification.
- Statistical significance claimed but not substantiated with tests or uncertainty.
- Ambiguity in the two-stage method vs experimental use of MTP/SEDD; L_{mix} impact not evidenced.
- Several tables lack numbering/captions; some duplication across sections.

 Scores

Originality: 2

Quality: 2

Clarity: 2

Significance: 2

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 4

 AI Review from SafeReviewer will be automatically processed **Summary**

This paper explores the enhancement of algorithmic creativity in text generation by comparing and combining different decoding methods. The authors investigate next-token prediction (NTP), multi-token teacherless prediction (MTP), and discrete diffusion (SEDD) on four combinatorial tasks: Sibling Discovery, Triangle Discovery, Circle Construction, and Line Construction. The core idea is to move beyond the limitations of traditional NTP, which often leads to repetitive and uncreative outputs, by introducing methods that encourage more diverse and less memorized generations. The authors introduce seed-conditioning, a technique that prepends random prefixes to the input, and temperature scaling to control the level of stochasticity in the generation process. They also propose an alignment loss to maintain semantic consistency between restricted and adaptive prompts. The empirical findings suggest that MTP and SEDD, especially when combined with seed-conditioning, outperform NTP in terms of diversity and memorization, while maintaining a reasonable level of coherence. The paper's significance lies in its attempt to address the trade-off between coherence and creativity in text generation, offering a potential path towards more innovative and less predictable models. However, the paper's presentation and methodological choices raise several concerns that need to be addressed to fully realize its potential.

 Strengths

I found the paper's exploration of different decoding methods for enhancing creativity in text generation to be a valuable contribution. The comparison of NTP, MTP, and SEDD on minimal combinatorial tasks provides a clear illustration of the trade-offs between coherence, memorization, and diversity. The authors' attempt to move beyond the limitations of traditional NTP, which often leads to repetitive outputs, is a significant step forward. The introduction of seed-conditioning, while not entirely novel, is a practical approach to injecting randomness into the generation process. The empirical results, particularly the reduction in memorization rates with MTP and SEDD, are compelling and suggest that these methods hold promise for generating more creative and less predictable text. The paper's focus on minimal tasks allows for a clear understanding of the core mechanisms at play, which is beneficial for isolating the effects of the different methods. The authors' explicit definition of their goals, including maximizing creative output, minimizing memorization, and maintaining diversity, provides a solid foundation for their experimental design. The inclusion of an alignment loss to maintain semantic consistency is also a positive aspect, as it acknowledges the importance of preserving meaning while encouraging creativity. Overall, the paper's exploration of these techniques and their potential to enhance algorithmic creativity is a noteworthy contribution to the field.

 Weaknesses

After a thorough examination of the paper, I've identified several significant weaknesses that impact its overall quality and the reliability of its conclusions. First, the paper suffers from a lack of clarity and precision in its writing. There are numerous instances of typos, grammatical errors, and awkward phrasing, which make it difficult to follow the authors' arguments. For example, the abstract contains errors such as "challanges" instead of "challenges" and "wichwequantisC#t S" instead of "which we quantify as C#t S". These errors, along with others throughout the paper, detract from the paper's credibility and make it challenging to understand the proposed methods and results. My analysis confirms that these errors are not isolated incidents but rather a pervasive issue throughout the manuscript. Second, the paper's presentation of mathematical notation is inconsistent and often lacks clear definitions. For instance, the notation 'D' is used on line 18 without prior definition, and the notation 'E' is used in Equation 3 without a clear definition within the equation itself. While the text surrounding Equation 3 implies that 'E' represents a text encoder, this should be explicitly stated within the equation or its immediate description. This lack of clarity makes it difficult to understand the mathematical formulations and their implications. My verification confirms that these are not isolated instances, as several other mathematical notations are also introduced without proper definitions. Third, the paper's experimental setup raises concerns about the fairness of the comparison between NTP, MTP, and SEDD. The paper does not provide sufficient details on the implementation of each method, making it difficult to assess whether they were given comparable resources and optimization. The fact that MTP and SEDD are diffusion-based methods, which typically require more computational resources, is not adequately addressed. This lack of transparency makes it difficult to determine whether the observed performance differences are due to the methods themselves or to differences in their implementation or computational resources. My analysis of the methods section confirms that the paper lacks the necessary details to ensure a fair comparison. Fourth, the paper's use of seed-conditioning, while potentially effective, is not presented as a novel contribution. The paper does not adequately discuss the relationship between seed-conditioning and existing techniques such as gisting or other forms of controlled generation. The paper also fails to cite relevant works that have used similar techniques. This lack of contextualization makes it difficult to assess the true novelty and impact of the proposed method. My review of the related work section confirms that the paper does not adequately address the existing literature on similar techniques. Fifth, the paper's evaluation metrics, while standard, are not always clearly defined or justified. For example, the diversity metric is defined as the number of unique outputs divided by the total number of outputs, but the paper does not discuss the limitations of this metric or its suitability for evaluating creative text generation. The paper also does not provide a clear justification for the specific values chosen for the hyperparameters, such as the alpha values for each method. This lack of transparency makes it difficult to assess the validity of the experimental results. My analysis of the experimental setup confirms that the paper lacks a thorough discussion of the evaluation metrics and hyperparameter choices. Finally, the paper's overall structure and organization are somewhat disjointed, with concepts and definitions introduced out of order. This makes it difficult to follow the authors' line of reasoning and understand the relationships between the different components of their proposed framework. My overall impression is that the paper needs significant revision to address these issues and improve its clarity, rigor, and overall quality.

Suggestions

To address the identified weaknesses, I recommend the following concrete actions. First, the authors should thoroughly revise the paper to correct all typos, grammatical errors, and awkward phrasing. This includes ensuring that all mathematical notations are defined before their first use and that all equations are clearly explained. The authors should also ensure that the paper is written in a clear and concise style, avoiding jargon and technical terms that are not essential to the discussion. Second, the authors should provide a more detailed description of the implementation of each method, including the specific architectures used, the optimization techniques, and the computational resources allocated. This will help to ensure a fair comparison between NTP, MTP, and SEDD and allow readers to assess the validity of the experimental results. Third, the authors should provide a more thorough discussion of the relationship between seed-conditioning and existing techniques, such as gisting and other forms of controlled generation. This should include a review of the relevant literature and a clear explanation of how seed-conditioning differs from these existing methods. The authors should also provide a more detailed explanation of the mechanism by which seed-conditioning is applied and its impact on the generation process. Fourth, the authors should provide a more thorough discussion of the evaluation metrics used in the paper, including their limitations and their suitability for evaluating creative text generation. The authors should also provide a clear justification for the specific values chosen for the hyperparameters, such as the alpha values for each method. This should include a discussion of the sensitivity of the results to these parameters. Fifth, the authors should reorganize the paper to ensure that concepts and definitions are introduced in a logical order. This includes ensuring that all necessary background information is provided before introducing the proposed methods and results. The authors should also ensure that the paper flows smoothly and that the relationships between the different sections are clear. Finally, the authors should consider conducting additional experiments to further validate their findings and explore

the limitations of their proposed methods. This could include experiments on more complex tasks or with different types of data. By addressing these issues, the authors can significantly improve the quality and impact of their paper.

 Questions

Based on my analysis, I have several questions that I believe are critical to understanding the paper's contributions and limitations. First, regarding the experimental setup, I'm curious about the specific architectures used for NTP, MTP, and SEDD. Were these identical transformer architectures, or were there differences that might affect the comparison? Furthermore, what specific optimization techniques were used for each method, and how were the computational resources allocated? Second, concerning the seed-conditioning, I'm interested in understanding the precise mechanism by which the random prefixes influence the generation process. How does the model interpret these prefixes, and how does this interpretation lead to more diverse outputs? Also, what is the impact of different seed lengths and randomness on the generated outputs? Third, regarding the evaluation metrics, I'm curious about the choice of the diversity metric. Given that the outputs are relatively short and constrained, is this metric sufficient to capture the nuances of creative text generation? What other metrics could be used to provide a more comprehensive evaluation? Fourth, regarding the hyperparameter choices, I'm interested in understanding the rationale behind the specific values chosen for the alpha parameters for each method. How were these values determined, and how sensitive are the results to changes in these parameters? Finally, regarding the overall approach, I'm curious about the authors' perspective on the limitations of their proposed methods. What are the potential challenges in scaling these methods to more complex tasks or larger datasets? What future research directions do the authors envision to address these limitations? These questions are crucial for a deeper understanding of the paper's contributions and limitations, and I believe that addressing them would significantly enhance the paper's overall quality.

 Scores

Soundness: 2.0

Presentation: 1.0

Contribution: 1.75

Rating: 2.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights